OL | Organic Letters

pubs.acs.org/OrgLett Letter

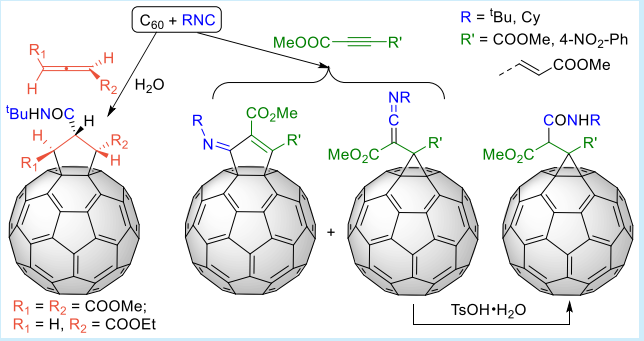
Isocyanide-Induced Annulation Leading to Cyclopento-, Methano-, and Cyclopentano-[60]Fullerene Derivatives

Yanbang Li, William P. Kopcha, Thomas J. Emge, Yue Sun, and Jianyuan Zhang*

Cite This: *Org. Lett.* 2021, 23, 8867–8872 Read Online

ACCESS | Metrics & More | Article Recommendations | Supporting Information

ABSTRACT: The three-component annulation reactions of C_{60}, alkyl isocyanide, and dimethyl acetylenedicarboxylate (DMAD) or unsymmetric alkynes are investigated to afford cyclopent-2-en-1-imino- and ketenimine methano-[60]fullerene derivatives, which, upon hydration in the presence of acid, yield the corresponding fullerene amides. Dimethyl 2,3-pentadienedioate, the allene counterpart of DMAD, and ethyl buta-2,3-dienoate undergo four-component annulation with C_{60}, alkyl isocyanide, and water under similar conditions to yield cyclopentano-[60]fullerene derivatives with similar amide groups.



Downloaded via FUDAN UNIV on January 17, 2022 at 10:29:06 (UTC).
See https://pubs.acs.org/sharingguidelines for options on how to legitimately share published articles.

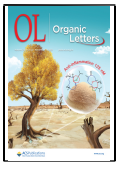
Fullerenes are promising building blocks in biomedicine,[1] organic and perovskite solar cells,[2] (photo)electrocatalytic systems,[3] and simple structural scaffolds for functional materials[4] due to their physicochemical, electrochemical, and photophysical properties and structural characteristics. Regioselective functionalization reactions of fullerenes have been actively sought after to improve the synthetic yield and the structural diversity of fullerene derivatives to realize their potential.[5]

Multicomponent reactions involving alkyl isocyanide and sp-hybridized carbon reactants (i.e., alkynes,[6] allenes[7]) have been widely used to construct cyclic structures in organic synthesis. These reactions are underexplored for the functionalization of fullerenes. The sole example, which is among an alkyl (cyclohexyl or *t*-butyl) isocyanide, a special alkyne dimethyl acetylenedicarboxylate (DMAD), and C_{60}, was reported to give two C_{60} derivatives: [3 + 2] cycloadduct **1** and [2 + 2] cycloadduct **2** (Scheme 1a).[8] However, even in this report, the structural elucidation of the derivatives left major questions. Recently, we reported the multicomponent reaction of a metallofullerene $Lu_3N@I_h$-C_{80}, *t*-butyl isocyanide (^{t}BuNC), and terminal or internal alkynes.[9] When DMAD was used as the alkyne, metallofulleroid **3** (Scheme 1a) with a ketenimine group formed, which was unambiguously confirmed by a single-crystal study. This result was very different from the reported C_{60} reactions. To fully understand the DMAD reaction and, more importantly, the substrate scope of reaction regarding other alkynes and allene reactants, we performed a systematic study of multicomponent reactions among ^{t}BuNC, a series of sp carbon reagents, and C_{60}. We found that the reaction of DMAD, C_{60}, and alkyl isocyanide would afford traditional [3 + 2] products similar to **1** and [1 + 2] adducts similar to **3**, as was confirmed by crystal data. Three-component reactions involving unsymmetric alkyne substrates that meet certain criteria could also give similar results. Additionally, the four-component reactions of allene substrates, dimethyl 2,3-pentadienedioate or ethyl buta-2,3-dienoate, C_{60}, ^{t}BuNC, and water yielded cyclopentano-[60]fullerene derivatives.

The reaction was first carried out with C_{60}, ^{t}BuNC, and DMAD in *o*-dichlorobenzene (Scheme 1b). Two products, later identified as the traditional [3 + 2] adduct **4a**[8] and ketenimine methanofullerene **5a**, respectively, were detected within a few minutes. The yields of **4a** and **5a** were strongly affected by the stoichiometry and the reaction time, as too high equivalents of the isocyanide and alkyne or a too long reaction time would lead to high-polarity mixtures that are insoluble in common solvents. Cyclohexyl isocyanide reacted in a similar fashion, yielding **4b** and **5b**. In both cases, the products **5** (**a** and **b**) were dominant. Furthermore, the hydrolysis of **5a** and **5b** under acidic conditions gave compounds **6a** and **6b** in high yields, whereas **5b** could gradually be converted to **6b** in air.

All of the products were purified by flash chromatography, and the spectroscopic data (Figures S1–S36) of **4**–**6** agree with the structures shown in Scheme 1b. Matrix-assisted laser-desorption ionization time-of-flight mass-spectrometry (MALDI-TOF MS) confirmed the molecular weights of **4**–

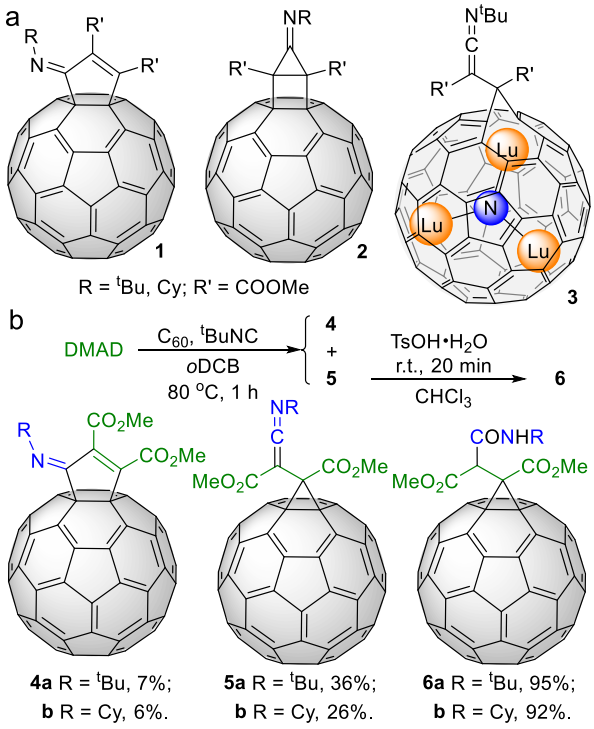
Received: October 4, 2021
Published: November 5, 2021



© 2021 American Chemical Society

https://doi.org/10.1021/acs.orglett.1c03371
Org. Lett. 2021, 23, 8867–8872

Scheme 1. (a) Reported Fullerene Derivatives from the Three-Component Reaction of Fullerenes, Alkyl Isocyanides, and DMAD and (b) This Work



6. All ^{1}H, ^{13}C, and $^{1}H-^{13}C$ heteronuclear multiple quantum coherence (HMQC) NMR spectra of **4a** and **4b** clearly pointed toward the 1,3-dipolar addition product (C_s symmetry), consistent with the previous report.[8] However, in theory, the position of the alkyl group can generate two stereoisomers, with R being cis or trans to the fullerene cage, which is an important aspect not addressed in the report.[8] Indeed, while we observed only one single isomer for **4a**, the ^{13}C NMR spectrum of **4b** showed a set of minor peaks of ~1/5 intensity due to the stereoisomerization. We hypothesized that the observed stereoisomer of **4a** and the major product in **4b** are the trans isomers because there is a large energetic cost for the bulky *t*-butyl group to be cis to C_{60} in the transition state, whereas such cost is less prohibitive for a flexible cyclohexyl group. The structures of **5a** and **5b** were also elucidated based on NMR. Taking **5a** for example, the ^{13}C NMR spectrum (Figure S9) shows C_s symmetry with only one peak (74.5 ppm) for two sp^3 carbon atoms on the cage, which suggests two functional groups being perpendicular to the reacted bond on the cage, such as a bridgehead carbon atom in methanofullerenes or fulleroids. On the basis of additional 1D and 2D NMR spectra (Figures S11–S13), the ketenimine methanofullerene is the most likely product, similar to **3**,[9] except that the C_{60} derivative is a closed-cage methanofullerene instead of an open-cage fulleroid based on the chemical shift of two sp^3 carbon atoms on the cage (74.5 ppm for **5a**, 74.2 ppm for **5b**). With the structures of **5a** and **5b**, the structures of hydration products **6a** and **6b** can be easily derived with the MS and 1D and 2D NMR characterizations.

To test our hypothesis on **4** and to confirm our structural elucidation of **5**, we performed single-crystal analysis on **4a** and **5a** (Figure 1, Figures S6 and S14). For **4a**, the five-membered ring at the [6,6] position was verified with a clear trans

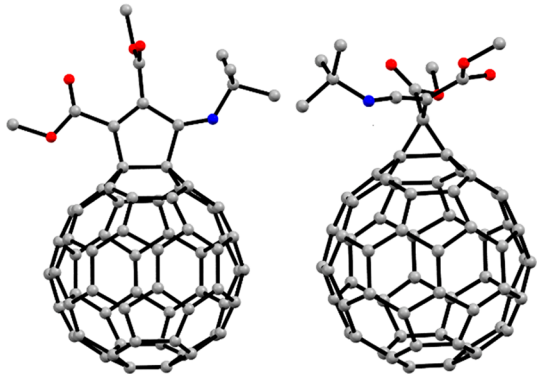


Figure 1. Single-crystal X-ray structure of compounds **4a** (left) and **5a** (right). Hydrogen atoms are not shown.

relationship between the fullerene cage and the *t*-butyl group, as was predicted. The ketenimine methanofullerene structure of **5a** was also definitively confirmed. On the basis of the high similarity of **5a** and **5b** in almost every way (various 1D and 2D NMR spectra, polarity, etc.), we expect **5b** to have the same ketenimine group with a different substituent.

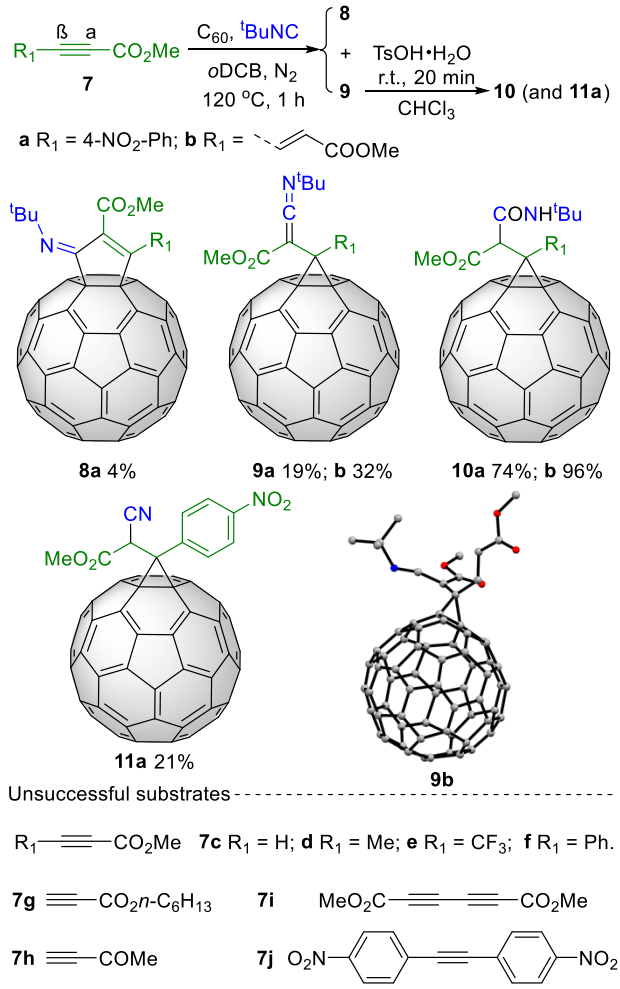
Given that our products **5** are different from the literature, we repeated our reaction using ^{t}BuNC in toluene, the same solvent used in the original report,[8] to exclude the possibility of the solvent effect resulting in different products. Still, the same products **4a** and **5a** were obtained. A careful examination of the ^{13}C NMR spectrum revealed that our spectrum of **5a** is identical to the reported spectrum of **2**, except with higher signal-to-noise (Figure S10 shows the comparison of the two spectra). Although structure **2** was a plausible assignment at the time, both spectra are more consistent with structure **5a**. More importantly, the structure of **5a** was confirmed by crystal analysis, which was not performed in the previous report.[8] Therefore, we conclude that the same product was misrepresented in the report 16 years ago given the sample amount and experimental conditions back then. Correspondingly, the correct hydration products were shown as **6** in the scheme based on the structures of **5** and their NMR data.

To further understand the reaction, we then set out to evaluate the scope of other alkyne reactants, as shown in Scheme 2. First, with one methyl ester group of DMAD replaced, **7a**[10] and **7c–f** as well as additional terminal alkynes **7g, h** were used as reactants and optimized with a range of stoichiometry and reaction temperatures. Only **7a** afforded similar products **8a** and **9a,** as in the DMAD reactions (Figures S37–S48), which indicates the importance of an electron-withdrawing group (EWG) on a phenyl ring, such as the 4-nitrophenyl group. During the process, ^{t}BuNC attacked the *α*-C of **7a** to generate the reactive 1,3- or 1,4-dipolar species,[9] which reacted with C_{60} to yield **8a** and **9a**. The attacking position, *α*-C, is proved by the $^{1}H-^{13}C$ heteronuclear multiple bond coherence (HMBC) 2D NMR spectrum of **9a** (Figure S48), in which the aromatic protons interact with the newly formed bridge carbon, whereas the methyl protons of the ester group do not. In the next step, the hydration of **9a** under the same acidic conditions gave amide **10a** (Figures S56–S61) and another byproduct **11a** (Figures S62–S67) with a newly formed cyano group.

The formation of these two compounds is due to the usual transformation of the ketenimine group[11] in **9a**. Encouraged by the function of the 4-nitrophenyl group, we synthesized

https://doi.org/10.1021/acs.orglett.1c03371
Org. Lett. 2021, 23, 8867–8872

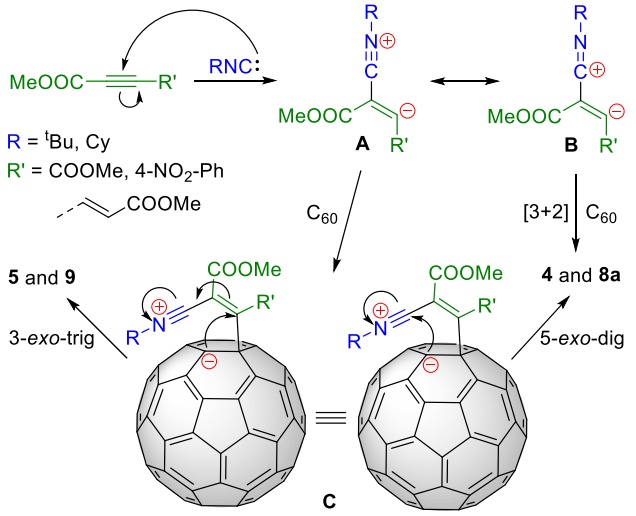
Scheme 2. Reactions Using Unsymmetric Alkyne Substrates and the Structure of the Products



compound **7j**[12] with two 4-nitrophenyl groups at both ends of the C≡C triple bond, but it failed to give similar results, which showed the key role of the ester group in stabilizing the α carbanion intermediate. In addition, **7b**[13] with a C=C double bond conjugated with the C≡C triple bond, used in the functionalization of C_{60} mediated by PPh_3,[13] was synthesized and used in the ^{t}BuNC-induced reaction of C_{60}. Compound **9b**, with the C=C double bond remaining, was obtained as the only isolable product (Figures S49−S55), which can be converted into amide **10b** (Figures S68−S73) in the presence of acid. The crystallization of these products was very challenging, and after extensive optimization, we were only able to get a decent crystal for **9b**, which is shown in Scheme 2 and Figure S55. When the double bond in **7b** was changed to a triple bond, **7i**[14] yielded a similar product in trace amount after a much longer reaction time at a much higher temperature. In summary, this reaction works when R_1 provides both an electron-withdrawing inductive effect and a resonance effect that stabilize the anionic intermediate.

The proposed mechanism for the formation of methano- and cyclopropeno-fullerene derivatives is proposed in Scheme 3. Nucleophilic attacks of alkyl isocyanide on alkynes yield zwitterionic species **A** and **B**. The concerted [3 + 2] cycloaddition of **B** and C_{60} generates typical [3 + 2] products **4** and **8a**. Carbanion **A** reacts with C_{60} to give fullerene anion **C**, which could undergo two cyclization pathways: 5-*exo*-dig

Scheme 3. Proposed Mechanism of the Formation of Methano- and Cyclopenteno-fullerenes



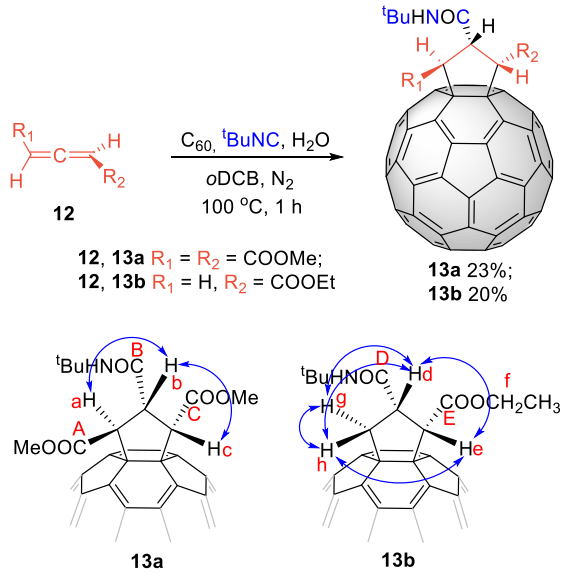
and 3-*exo*-trig. The 5-*exo*-dig ring-closure of **C** also affords [3 + 2] products **4** and **8a**, whereas the 3-*exo*-trig ring closure forms ketenimine derivatives **5** and **9**. The ketenimine derivatives have significantly higher yields than the [3 + 2] products, indicating that the 3-exo-trig ring closure is favorable over the 5-*exo*-dig process in this case. The mechanism is similar to that in our recent report on the $Lu_3N@C_{80}$ reaction,[9] except that the [3 + 2] adduct is formed in an appreciable amount and the rearrangement of methanofullerene forming open-cage fulleroids did not happen.

Another form of sp carbon atoms, cumulated alkenes, can undergo cycloadditions with C_{60} induced by phosphines.[15] Although a multicomponent reaction involving alkyl isocyanide and allenoates has been widely used to construct cyclic structures in organic synthesis,[7] such a reaction has never been applied in the functionalization of fullerene to the best of our knowledge. Replacing the triple bond with cumulated double bonds, we performed the reaction of C_{60}, ^{t}BuNC, and the allenoate counterpart of DMAD. Dimethyl 2,3-pentadienedioate **12a**[16] reacted with C_{60} and ^{t}BuNC and yielded only one isolable product, later identified as the **13a** (Figures S74−S79), with water involved in this process. Ethyl buta-2,3-dienoate **12b** underwent a similar four-component annulation to form **13b** (Figures S80−S85).

MALDI-TOF spectra (Figures S74 and S80) showed molecular ion signals for **13a,b** as the sum of all reactants plus one molecule of water. The ^{1}H, ^{13}C, DEPT-135, and HMQC NMR spectra (Figures S75−S79 and S81−S85) of **13a** and **13b** showed the formation of the 1,3-dipolar addition products (C_s symmetry). ^{1}H and HMBC NMR spectra reveal more detailed stereochemistry regarding the relative positions of the amide and ester groups (Scheme 4). For **13a**, the orientations of three special protons were determined as follows: (1) The chemical shifts of H_a and H_c were significantly different. (2) There were two different coupling constants between H_b and H_a ($^3J_{trans} \approx 12.8$ Hz) or H_c ($^3J_{cis} = 6.8$ Hz). (3) In HMBC, H_c interacts with C_C, whereas H_a and H_b interact with all three of the carbonyl carbons. Therefore, H_a and H_c are nonequivalent; that is, they have different relative positions with H_b. The situation of **13b** is further complicated by the greater number of protons. The geminal coupling between H_g and H_h ($^2J \approx 12.3$ Hz) confirms the formation of

https://doi.org/10.1021/acs.orglett.1c03371
Org. Lett. 2021, 23, 8867−8872

Scheme 4. Four-Component Reaction Involving Allenoates



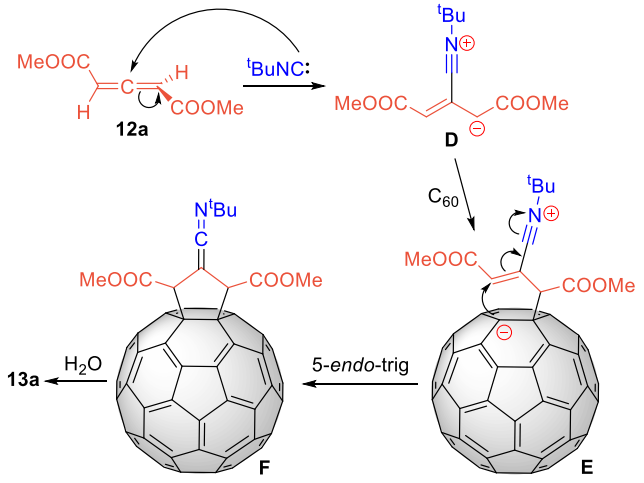
	chemical shift/ppm	coupling with	HMBC interaction with carbonyl carbons
H_a	5.87	H_b (J = 12.7 Hz)	C_A, C_B, C_C
H_b	4.60	H_a (J = 12.8 Hz) H_c (J = 6.8 Hz)	C_A, C_B, C_C
H_c	4.88	H_b (J = 6.8 Hz)	C_C
H_d	4.21-4.35 including H_f	H_e (J = 6.6 Hz) H_g (J = 13.3 Hz) H_h (J = 6.2 Hz)	C_D, C_E
H_e	4.85	H_d (J = 6.6 Hz) H_h (J = 1.2 Hz)	C_E
H_g	4.72	H_d (J = 13.3 Hz) H_h (J = 12.2 Hz)	C_D
H_h	3.62	H_d (J = 6.2 Hz) H_e (J = 1.3 Hz) H_g (J = 12.4 Hz)	N/A

the methylene group. The long-range coupling between H_h and H_e ($J \approx 1.2$ Hz) determines the cis orientations of these two protons. The orientation of H_d is based on the following points: (1) There are three coupling constants between H_d and H_e ($^3J_{cis} = 6.6$ Hz), H_g ($^3J_{trans} = 13.3$ Hz), or H_h ($^3J_{cis} = 6.2$ Hz) respectively. (2) In HMBC, among H_e, H_g, and H_h, only H_g interacts with amide C_D due to the cis orientations of H_g and C_D.

On the basis of the literature[7,15] and our results,[9] we have proposed the mechanism for the formation of **13**, using **13a** as an example (Scheme 5). ^{t}BuNC reacts with **12a** to generate intermediate **D**; then, the anionic carbon atom attacks the [6,6] bond of C_{60} to form the fullerene carbanion **E**, which can undergo two cyclization pathways: 5-*exo*-dig and 5-*endo*-trig. In this case, the favored 5-*endo*-trig ring closure of species **E** gives species **F** with a ketenimine group. Similar to the hydration of **5** and **9** under acidic conditions, species **F** directly reacts with water to afford compound **13a**.

In summary, the three-component cycloaddition reaction of C_{60}, alkyl isocyanide, and DMAD is investigated to afford cyclopent-2-en-1-imino- and ketenimine methano-[60]-fullerene derivatives, as confirmed by single-crystal analysis. Hydration of the ketenimine methanofullerene in the presence

Scheme 5. Proposed Mechanism of the Formation of Cyclopentanofullerenes Exemplified with 13a



of acid gives the corresponding fullerene amides. Additional versions of alkynes have been tested as reactants under similar conditions, which suggests that the reaction requires an ester group on one side and a substituent providing both inductive and resonance effects on the other side of the triple bond. Furthermore, the four-component reaction of C_{60}, alkyl isocyanide, allenoates, and water is also investigated and provides cyclopentano-[60]fullerene derivatives. The multi-component annulation reaction involving allenoates will be explored with other carbon cages, such as C_{70} and endohedral fullerenes.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.1c03371.

Experimental procedures, spectroscopic data, and copies of ^{1}H and ^{13}C NMR spectra for all new compounds (PDF)

Accession Codes

CCDC 2047096, 2047102, and 2112413 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

■ AUTHOR INFORMATION

Corresponding Author

Jianyuan Zhang − *Department of Chemistry and Chemical Biology, Rutgers, The State University of New Jersey, Piscataway, New Jersey 08854, United States;* orcid.org/0000-0002-2919-2734; Email: jy.zhang@rutgers.edu

Authors

Yanbang Li − *Department of Chemistry and Chemical Biology, Rutgers, The State University of New Jersey, Piscataway, New Jersey 08854, United States*

https://doi.org/10.1021/acs.orglett.1c03371
Org. Lett. 2021, 23, 8867−8872

William P. Kopcha − *Department of Chemistry and Chemical Biology, Rutgers, The State University of New Jersey, Piscataway, New Jersey 08854, United States*
Thomas J. Emge − *Department of Chemistry and Chemical Biology, Rutgers, The State University of New Jersey, Piscataway, New Jersey 08854, United States*
Yue Sun − *Department of Chemistry and Chemical Biology, Rutgers, The State University of New Jersey, Piscataway, New Jersey 08854, United States*

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.orglett.1c03371

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work is supported by the Department of Energy of USA (grant no. DE-SC0020260).

■ REFERENCES

(1) For recent reviews, see: (a) Nierengarten, J.-F. Fullerene Hexa-adduct Scaffolding for the Construction of Giant Molecules. *Chem. Commun.* **2017**, *53*, 11855. (b) Castro, E.; Garcia, A. H.; Zavala, G.; Echegoyen, L. Fullerenes in Biology and Medicine. *J. Mater. Chem. B* **2017**, *5*, 6523. (c) Harano, K.; Nakamura, E. Interfacial Chemistry of Conical Fullerene Amphiphiles in Water. *Acc. Chem. Res.* **2019**, *52*, 2090.

(2) For recent reviews, see: (a) Gatti, T.; Menna, E.; Meneghetti, M.; Maggini, M.; Petrozza, A.; Lamberti, F. The Renaissance of Fullerenes with Perovskite Solar Cells. *Nano Energy* **2017**, *41*, 84. (b) Umeyama, T.; Imahori, H. Isomer Effects of Fullerene Derivatives on Organic Photovoltaics and Perovskite Solar Cells. *Acc. Chem. Res.* **2019**, *52*, 2046. (c) Deng, L.-L.; Xie, S.-Y.; Gao, F. Fullerene-Based Materials for Photovoltaic Applications: Toward Efficient, Hysteresis-Free, and Stable Perovskite Solar Cells. *Adv. Electron. Mater.* **2018**, *4*, 1700435. (d) Cui, C.; Li, Y.; Li, Y. Fullerene Derivatives for the Applications as Acceptor and Cathode Buffer Layer Materials for Organic and Perovskite Solar Cells. *Adv. Energy Mater.* **2017**, *7*, 1601251. (e) Castro, E.; Murillo, J.; Fernandez-Delgado, O.; Echegoyen, L. Progress in Fullerene-based Hybrid Perovskite Solar Cells. *J. Mater. Chem. C* **2018**, *6*, 2635. (f) Jia, L.; Chen, M.; Yang, S. Functionalization of Fullerene Materials toward Applications in Perovskite Solar Cells. *Mater. Chem. Front.* **2020**, *4*, 2256.

(3) For recent reviews, see: (a) Lu, F.; Neal, E. A.; Nakanishi, T. Self-Assembled and Nonassembled Alkylated-Fullerene Materials. *Acc. Chem. Res.* **2019**, *52*, 1834. (b) Puente Santiago, A. R.; Fernandez-Delgado, O.; Gomez, A.; Ahsan, M. A.; Echegoyen, L. Fullerenes as Key Components for Low-Dimensional (Photo) electrocatalytic Nanohybrid Materials. *Angew. Chem., Int. Ed.* **2021**, *60*, 122. (c) Zieleniewska, A.; Lodermeyer, F.; Roth, A.; Guldi, D. M. Fullerenes−How 25 Years of Charge Transfer Chemistry Have Shaped Our Understanding of (Interfacial) Interactions. *Chem. Soc. Rev.* **2018**, *47*, 702. (d) Collavini, S.; Delgado, J. L. Fullerenes: The Stars of Photovoltaics. *Sustainable Energy Fuels* **2018**, *2*, 2480. (e) Megiatto, J. D., Jr; Guldi, D. M.; Schuster, D. I. Design, Synthesis and Photoinduced Processes in Molecular Interlocked Photosynthetic [60] Fullerene Systems. *Chem. Soc. Rev.* **2020**, *49*, 8.

(4) Hirsch, A.; Brettreich, M. *Fullerenes: Chemistry and Reactions*; Wiley-VCH Verlag GmbH & Co. KGaA: Weinheim, Germany, 2005.

(5) For recent examples, see: (a) Zhou, Z.; Han, H.; Chen, Z.; Gao, R.; Liu, Z.; Su, J.; Xin, N.; Yang, X.; Gan, L. Concise Synthesis of Open-Cage Fullerenes for Oxygen Delivery. *Angew. Chem., Int. Ed.* **2019**, *58*, 17690. (b) Hashikawa, Y.; Li, H.; Murata, Y. Reactions of C_{60} with Pyridazine and Phthalazine. *Chem. - Eur. J.* **2021**, *27*, 7507. (c) Wachter, M.; Jurkiewicz, L.; Hirsch, A. Sequential Tether-Directed Synthesis of New [3:2: 1] Hexakis-Adducts of C_{60} with a Mixed Octahedral Addition Pattern. *Chem. - Eur. J.* **2021**, *27*, 7677. (d) Ubasart, E.; Borodin, O.; Fuertes-Espinosa, C.; Xu, Y.; Garcia-Simon, C.; Gomez, L.; Juanhuix, J.; Gandara, F.; Imaz, I.; Maspoch, D.; von Delius, M.; Ribas, X. A Three-shell Supramolecular Complex Enables the Symmetry-mismatched Chemo-and Regioselective Bis-functionalization of C_{60}. *Nat. Chem.* **2021**, *13*, 420. (e) Leonhardt, V.; Fimmel, S.; Krause, A.-M.; Beuerle, F. A Covalent Organic Cage Compound Acting as a Supramolecular Shadow Mask for the Regioselective Functionalization of C_{60}. *Chem. Sci.* **2020**, *11*, 8409. (f) Lu, W.-Q.; Zhou, D.-B.; Yin, Z.-C.; Liu, Q.-S.; Wang, G.-W. A Copper-promoted Synthesis of Epoxy-bridged [60]Fullerene-fused Lactones and Further Derivatization. *Chem. Commun.* **2021**, *57*, 7043. (g) Liu, Z.; Yin, Z.-C.; Lu, W.-Q.; Niu, C.; Chen, M.; Yang, S.; Wang, G.-W. Cu (I)-Catalyzed Synthesis of [60] Fullerene-Fused Lactams and Further Electrochemical Functionalization. *Org. Lett.* **2021**, *23*, 4051. (h) Wu, C.; Liu, T.-X.; Zhang, P.; Zhu, X.; Zhang, G. Iron-Catalyzed Redox-Neutral Radical Cascade Reaction of [60]Fullerene with γ, δ-Unsaturated Oxime Esters: Preparation of Free (N-H) Pyrrolidino[2′,3′:1,2]fullerenes. *Org. Lett.* **2020**, *22*, 7327. (i) Ma, J.; Liu, T.-X.; Zhang, P.; Zhao, X.; Zhang, G. Metal-Free-Catalyzed Three-Component [2 + 2+2] Annulation Reaction of [60]Fullerene, Ketones, and Indoles: Access to Diverse [60]Fullerene-Fused 1,2-Tetrahydrocarbazoles. *Org. Lett.* **2021**, *23*, 1775. (j) Ma, J.; Liu, T.-X.; Zhang, P.; Zhang, C.; Zhang, G. Palladium-catalyzed Domino Spirocyclization of [60]Fullerene: Synthesis of Diverse [60]-Fullerene-fused Spiro[4,5]/[5,5] Derivatives. *Chem. Commun.* **2021**, *57*, 49. (k) Malliaros, N. G.; Kellner, I. D.; Drewello, T.; Orfanopoulos, M. Decatungstate-Photocatalyzed Addition of Lactones to [60]Fullerene. *J. Org. Chem.* **2021**, *86*, 9876. (l) Artigas, A.; Castanyer, C.; Roig, N.; Lledó, A.; Solà, M.; Pla-Quintana, A.; Roglans, A. Synthesis of Fused Dihydroazepine Derivatives of Fullerenes by a Rh-Catalyzed Cascade Process. *Adv. Synth. Catal.* **2021**, *363*, 3835.

(6) (a) Gulevich, A. V.; Zhdanko, A. G.; Orru, R. V. A.; Nenajdenko, V. G. Isocyanoacetate Derivatives: Synthesis, Reactivity, and Application. *Chem. Rev.* **2010**, *110*, 5235. (b) Sadjadi, S.; Heravi, M. M.; Nazari, N. Isocyanide-based Multicomponent Reactions in the Synthesis of Heterocycles. *RSC Adv.* **2016**, *6*, 53203. (c) Li, J.; Liu, Y.; Li, C.; Jie, H.; Jia, X. Atom-economical Synthesis of the Functionalized Spirocyclic Oxindole-butenolide via Three-component [2+ 2+ 1] Cycloaddition Strategy. *Green Chem.* **2012**, *14*, 1314. (d) Jie, H.; Li, J.; Li, C.; Jia, X. Isocyanide-Based Multicomponent [2+ 2+ 1]-Cycloaddition Strategy to Construct Functionalized Spirocyclic Oxindoles. *Synlett* **2012**, *23*, 2274.

(7) Selected examples: (a) Li, J.; Liu, Y.; Li, C.; Jia, X. Silver Hexafluoroantimonate-catalyzed Three-component [2+ 2+ 1] Cycloadditions of Allenoates, Dual Activated Olefins, and Isocyanides. *Adv. Synth. Catal.* **2011**, *353*, 913. (b) Li, J.; Liu, Y.; Li, C.; Jia, X. Syntheses of Spirocyclic Oxindole-Butenolides by Using Three-Component Cycloadditions of Isocyanides, Allenoates, and Isatins. *Chem. - Eur. J.* **2011**, *17*, 7409. (c) Li, J.; Wang, N.; Li, C.; Jia, X. Multicomponent Reaction to Construct Spirocyclic Oxindoles with a Michael (Triple Michael)/Cyclization Cascade Sequence as the Key Step. *Chem. - Eur. J.* **2012**, *18*, 9645. (d) Jia, S.; Su, S.; Li, C.; Jia, X.; Li, J. Multicomponent Cascade Cycloaddition Involving Tropone, Allenoate, and Isocyanide: A Rapid Access to a 7, 6, 5-Fused Tricyclic Skeleton. *Org. Lett.* **2014**, *16*, 5604. (e) Tang, Z.; Liu, Z.; An, Y.; Jiang, R.; Zhang, X.; Li, C.; Jia, X.; Li, J. Isocyanide-Based Multicomponent Bicyclization with Substituted Allenoate and Isatin: Synthesis of Unusual Spirooxindole Containing [5.5]-Fused Heterocycle. *J. Org. Chem.* **2016**, *81*, 9158.

(8) Zhou, Z.; Magriotis, P. A. A New Method for the Functionalization of [60] Fullerene:An Unusual 1, 3-DipolarCycloaddition Pathway Leading to a C60 Housane Derivative. *Org. Lett.* **2005**, *7*, 5849.

(9) Li, Y. B.; Emge, T. J.; Moreno-Vicente, A.; Kopcha, W. P.; Sun, Y.; Mansoor, I. F.; Lipke, M. C.; Hall, G. S.; Poblet, J. M.; Rodríguez-Fortea, A.; Zhang, J. Unexpected Formation of Metallofulleroids from Multicomponent Reactions, with Crystallographic and Computational

Studies of the Cluster Motion. *Angew. Chem., Int. Ed.* **2021**, DOI: 10.1002/anie.202110881.

(10) (a) Pan, J.-L.; Xie, P.; Chen, C.; Hao, Y.; Liu, C.; Bai, H.-Y.; Ding, J.; Wang, L.-R.; Xia, Y.; Zhang, S.-Y. Rhodium(III)-Catalyzed Redox-Neutral Cascade [3 + 2] Annulation of N-Phenoxyacetamides with Propiolates via C-H Functionalization/Isomerization/Lactonization. *Org. Lett.* **2018**, *20*, 7131. (b) Skelly, P. W.; Sae-Jew, J.; Kitos Vasconcelos, A. P.; Tasnim, J.; Li, L.; Raskatov, J. A.; Braslau, R. Relative Rates of Metal-Free Azide-Alkyne Cycloadditions: Tunability over 3 Orders of Magnitude. *J. Org. Chem.* **2019**, *84*, 13615.

(11) Lu, P.; Wang, Y. The Thriving Chemistry of Ketenimines. *Chem. Soc. Rev.* **2012**, *41*, 5687.

(12) Ruengsangtongkul, S.; Chaisan, N.; Thongsornkleeb, C.; Tummatorn, J.; Ruchirawat, S. Rate Enhancement in CAN-Promoted $Pd(PPh_3)_2Cl_2$-Catalyzed Oxidative Cyclization: Synthesis of 2-Ketofuran-4-carboxylate Esters. *Org. Lett.* **2019**, *21*, 2514.

(13) (a) Chuang, S.-C.; Santhosh, K. C.; Lin, C.-H.; Wang, S.-L.; Cheng, C.-H. Synthesis and Chemistry of Fullerene Derivatives Bearing Phosphorus Substituents. Unusual Reaction of Phosphines with Electron-Deficient Acetylenes and C_{60}. *J. Org. Chem.* **1999**, *64*, 6664. (b) Chuang, S.-C.; Deng, J.-C.; Chan, F.-W.; Chen, S.-Y.; Huang, W.-J.; Lai, L.-H.; Rajeshkumar, V. [3 + 2] Cycloaddition of Dialkyl (E)-Hex-2-en-4-ynedioates to [60]Fullerene by Phosphane-Promoted Tandem $\alpha(\delta')$-Michael Additions. *Eur. J. Org. Chem.* **2012**, *2012*, 2606.

(14) Deng, J.-C.; Chuang, S.-C. Multicomponent Reactions of Phosphines, Diynedioates, and Aryl Aldehydes Generated Furans Appending Reactive Phosphorus Ylides through Cumulated Trienoates as Key Intermediates: A Phosphine α-Addition-δ-Evolvement of an Anion Pathway. *Org. Lett.* **2014**, *16*, 5792.

(15) (a) Shu, L.-H.; Sun, W.-Q.; Zhang, D.-W.; Wu, S.-H.; Wu, H.-M.; Xu, J.-F.; Lao, X.-F. Phosphine-Catalysed [3 + 2] Cycloadditions of Buta-2,3-dienoates with [60]Fullerene. *Chem. Commun.* **1997**, 79. (b) O'Donovan, B. F.; Hitchcock, P. B.; Meidine, M. F.; Kroto, H. W.; Taylor, R.; Walton, D. R. M. Phosphine-Catalysed Cycloaddition of Buta-2,3-dienoates and But-2-ynoates to [60]Fullerene. *Chem. Commun.* **1997**, 81. (c) Marco-Martínez, J.; Marcos, V.; Reboredo, S.; Filippone, S.; Martín, N. Asymmetric Organocatalysis in Fullerenes Chemistry: Enantioselective Phosphine-catalyzed Cycloaddition of Allenoates onto C_{60}. *Angew. Chem., Int. Ed.* **2013**, *52*, 5115. (d) Girón, R. M.; Reboredo, S.; Marco-Martínez, J.; Filippone, S.; Martín, N. Organocatalysis for New Chiral Fullerene-based Materials. *Faraday Discuss.* **2014**, *173*, 311.

(16) Katoh, T.; Noguchi, C.; Kimura, H.; Fujiwara, T.; Ichihashi, S.; Nishide, K.; Kajimoto, T.; Node, M. A Practical Improvement of Crystallization-induced Asymmetric Transformation of Allene-1,3-dicarboxylates. *Tetrahedron: Asymmetry* **2006**, *17*, 2943.

https://doi.org/10.1021/acs.orglett.1c03371
Org. Lett. 2021, 23, 8867−8872